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Business Requirement Document

July,22,2025



Business Requirements Document (BRD)

Centralized Process Automation

1. Executive Summary

Milki Food Complex operates across multiple locations with complex workflows involving procurement, warehouse management, sales, delivery, payment tracking, and employee coordination. The factory's growth and increasing complexity demands an integrated ERP system to replace fragmented and manual processes.

This BRD outlines the scope, functionality, and technical expectations for a full-featured **ERP** system that supports Inventory, Sales, Logistics, Finance, HR, Procurement, and CRM operations under a single, centralized platform.

2. Problem Statement

Despite Milki Food Complex's growing operations and product distribution across regions, the current system of managing warehouse inventories, product flows, orders, and payments is largely manual or fragmented, resulting in significant operational inefficiencies. The following key problem areas have been identified:

2.1. Lack of Centralized Warehouse Oversight

Each warehouse operates in a silo, with minimal or no integration with a central system. This causes:

- *Inability to view real-time stock levels across locations.*
- *Delayed decision-making due to lack of timely data.*
- *Difficulties balancing product quantities across warehouses to meet demand.*

2.2. Inefficient Inventory Management

Product inflows (incoming from production) and outflows (shipped to customers) are often recorded manually or via basic spreadsheets, leading to:

- *Data entry errors and duplication.*
- *Miscounted stock and discrepancies between physical and recorded inventory.*
- *Time-consuming reconciliations, especially during audits or restocking.*

2.3. Poor Visibility into Order Lifecycle

The order processing workflow lacks a **digitized trail from customer request to shipment**. As a result:

- Orders are sometimes delayed, lost, or fulfilled incorrectly.
- No structured tracking exists to view the status of an order (pending, dispatched, delivered).
- Coordination between sales, dispatch, and finance teams becomes complex and error-prone.

2.4. Absence of Structured Payment Tracking

There is no robust linkage between **shipment records and payments received**. This causes:

- Shipped products to remain unpaid without timely detection.
- Finance teams to rely on manual reconciliations to identify unpaid orders.
- Reduced cash flow visibility and potential revenue leakages.

2.5. Unstructured Product Categorization

Products are not systematically **classified** based on **type, packaging, location, or pricing**. This leads to:

- Confusion during selection or packaging for shipment.
- Inconsistent pricing and difficulty managing product variants.
- Challenges in offering product insights to stakeholders or clients.

2.6. Limited User Accountability and Workflow Definition

There is no well-defined digital workflow to manage role-based tasks. This causes or will causes:

- Overlaps and gaps in task ownership across teams.
- Difficulty auditing who did what and when.
- Low accountability and process delays.

2.7. Reporting Limitations

The absence of **real-time** dashboards and reports impacts:

- Timely operational and financial decision-making.
- Executive visibility on KPIs such as stock turnover, unpaid shipments, and regional performance.
- Strategic planning based on reliable data insights.

3. Business Objectives

- Centralize management of operational workflows across departments.
- Improve efficiency, traceability, and accountability in every process.
- Reduce manual work, duplication, and data silos.
- Improve employee and vendor engagement.
- Enable a performance-driven, real-time decision-making environment.

4. Scope of the ERP System

Module	Description
Inventory Management	Handle multi-warehouse product tracking, stock levels, and movement.
Sales & Order Management	End-to-end customer order handling and sales tracking.
Logistics & Shipment Tracking	Shipment planning, dispatch, and delivery confirmation.
Finance & Accounts Receivable	Payment tracking, outstanding invoices, and financial alerts.
Product Lifecycle & Categorization	Manage products by type, grade, pricing, and production stages.
Workflow & User Management	Role-based access, approval flows, and audit logs.
Dashboards & Reporting	KPI dashboards, visual insights, and exportable reports.
Human Resource Management (HRM)	Track employee profiles, attendance, roles, leave, and performance.
Procurement Management	Manage vendor sourcing, purchase requests, and goods receipt.
Customer Relationship Management (CRM)	Manage customer data, communications, complaints, and feedback loops.

5. Stakeholder Analysis

Stakeholder	Responsibilities
Executive Management	Strategic decision-making, policy enforcement.
Warehouse Staff	Register, receive, and dispatch inventory.
Sales & Marketing	Process orders, track customers, and manage communication.
Finance Department	Track payments, processing invoices, financial reports.
Procurement Officers	Vendor onboarding, purchase requisitions and order tracking.
HR Department	Employee records, attendance, leave and payroll.
Logistics Coordinators	Shipment and delivery planning.
System Admin / IT Team	Setup, configuration, security, and system maintenance.

6. Functional Requirements

6.1. Inventory Management

ID	Description
IM-601	Track real-time stock across warehouses.
IM-602	Log inbound and outbound product movements.
IM-603	Support batch IDs, expiry dates, and location bins.
IM-603	Trigger low stock alerts.

6.2. Sales & Order Management

ID	Description
SOM-201	Customer order creation and tracking.
SOM-202	Stock reservation per order.
SOM-203	Order status lifecycle: Draft → Confirmed → Shipped → Delivered → Paid.
SOM-204	Generate order documents (invoices, delivery slips).

6.3. Logistics & Shipment

ID	Description
LS-301	Dispatch planning with route and carrier assignment.
LS-302	Shipment status: Scheduled, In Transit, Delivered, Returned.
LS-303	Record proof of delivery and exceptions.

6.4. Finance & Accounts Receivable

ID	Description
FAR-401	Record payments by customer and order.
FAR-402	Generate aging reports and outstanding balances.
FAR-403	Alerts for unpaid or overdue invoices.
FAR-404	Link with shipment and sales modules.

6.5. Product Categorization & Lifecycle

ID	Description
PCL-501	Classify products by line (e.g., cut flowers, potted plants).
PCL-502	Define pricing, grading, and packaging rules.
PCL-503	Log production stages: harvesting, storage, dispatch.

6.6. Workflow & User Management

ID	Description
WU-601	Role-based access (Admin, Finance, HR, Logistics, etc.).
WU-602	Approval flows for high-value transactions or critical activities.
WU-603	Track user activities and audit logs.

6.7. Dashboards & Reports

ID	Description
DR-701	Visual dashboards per module (Inventory, Sales, HR, etc.).
DR-702	Custom filters for location, period, status.
DR-703	Scheduled reporting via email.
DR-704	Export to Excel/PDF.

6.8. Human Resource Management (HRM)

ID	Description
HR-801	Maintain employee master data (ID, name, role, contact, department).
HR-802	Manage attendance and work schedules.
HR-803	Record leave types, applications, and approvals.
HR-804	Handle onboarding/offboarding processes.
HR-805	Generate performance reports and payroll base data.

6.9. Procurement Management

ID	Description
PM-901	Register and manage approved suppliers.
PM-902	Create and approve purchase requisitions (PRs).
PM-903	Convert PRs to purchase orders (POs).
PM-904	Log goods receipt, quantity, and inspection status.
PM-905	Link purchases to financial and inventory modules.

6.10. Customer Relationship Management (CRM)

ID	Description
CRM-1001	Maintain customer profiles, contact history, and preferences.
CRM-1002	Manage feedback, complaints, and case resolution.
CRM-1003	Log customer interactions across sales, support, and delivery.
CRM-1004	Generate customer engagement reports.

7. Non-Functional Requirements

ID	Description
NFR-1	Web-based, mobile-responsive, and offline-capable (optional).
NFR-2	Secure login, 2FA, encryption for sensitive data.
NFR-3	Scalable microservices-ready architecture.
NFR-4	Multilingual support (Amharic/English).
NFR-5	Integration-ready via REST APIs (e.g., GPS, SMS, financial platforms).
NFR-6	Regular backups, high availability, and recovery readiness.

8. Data Model Summary (Entities)

Entity	Key Attributes
Product	ID, Name, Type, Price, Quantity, Grade
Warehouse	ID, Name, Location, Capacity
Customer	ID, Name, Phone, Email, Address
Order	ID, Customer ID, Products, Status
Shipment	ID, Carrier, Destination, Status
Payment	Order ID, Amount, Method, Status
Employee	ID, Name, Role, Department, Status
Vendor	ID, Name, Contact, Category
Requisition	ID, Requested Items, Status
Ticket (CRM)	ID, Customer ID, Type, Status, Resolution Notes

9. Workflow Overviews

Procurement Workflow:

Purchase Request → Approval → PO Creation → Vendor Fulfillment → Goods Receipt → Inventory Update → Payment

HR Workflow:

Employee Onboarding → Attendance Tracking → Leave Requests → Payroll Export

CRM Workflow:

Customer Issue Logged → Assigned to Team → Resolution → Feedback Captured

10. Assumptions

- Staff will be trained on new ERP workflows.
- Current manual processes will be gradually replaced.
- Necessary infrastructure (network, devices, support) will be provided.
- External systems (e.g., accounting software) may need integration later.

11. Future-Proofing

- Plug-in modules for export customs, compliance, or sustainability tracking.
- **AI-based inventory prediction or customer satisfaction analytics.**
- Integration with mobile field sales apps or handheld scanners.

12. Appendices

Appendix A: Access Matrix

Role	Inventor y	Sale s	Logistic s	Financ e	HR	Procuremen t	CR M	Admi n
Admin	Full	Full	Full	Full	Ful l	Full	Full	Full
Sales Staff	View	RW	View	View	-	-	RW	-
Warehouse Clerk	RW	View	RW	-	-	-	-	-
Finance Officer	View	View	View	RW	-	View	-	-
HR Officer	-	-	-	View	RW	-	-	-
Procuremen t Team	View	-	View	View	-	RW	-	-